Tobias Anuf

+595 (971) 862-558 | tobiasanuf@gmail.com | LinkedIn | github.com/tobiascca | tobiascca.github.io/personal-portfolio

Technical Skills

Programming Languages: Python, SQL, Java, C/C++, C#, Swift, JavaScript, HTML/CSS
ML & Data: scikit-learn, PyTorch, TensorFlow / TF Lite, OpenCV, MediaPipe, Transformers, Pandas, NumPy
Data & Tools: Amazon Redshift, MySQL, MongoDB, Git, Google Cloud Platform, Unity, Raspberry Pi, VS Code
Languages: Spanish (Native), English (Proficient / C2)

Education

Universidad Politécnica Taiwán-Paraguay — Asunción, PY
B.E. in Information Engineering — *Feb. 2023 – Present*

- ***Relevant coursework**: Computer Architecture, Operating Systems, Data Structures and Algorithms, Advanced Computer Networks, Digital Logic Design with Verilog*

National Taiwan University of Science and Technology (NTUST) — Taipei, TW
3-semester exchange — *Feb. 2025 – June 2026*

- ***Relevant coursework**: Machine Learning and Big Data Analysis, Computer Vision and Applications, Data Mining, Text Mining, Social Networks, Database Systems, Java Programming*

Experience

Data Science Intern — Aug. 2026 – Oct. 2026
Dato Solutions S.A. – Research & Development — *Asunción, PY*

- Built an ML system that recovers the importer tax ID that Paraguay's customs open data stopped publishing in 2024, restoring "who imports what" for the company's trade-intelligence product
- Benchmarked four approaches on 26M+ records; a TF-IDF + kNN model nearly doubled top-1 accuracy over the rule-based baseline (29% → 57%), reaching 65% top-1 / 80% top-5 as data scaled
- Set a calibrated confidence threshold (60% of shipments answered at 86.5% precision) and validated totals against Central Bank of Paraguay statistics; built a resumable Python + Redshift batch pipeline
- Fixed calculation errors in the company's LLM query assistant, rewrote its instructions, and built an automated validation suite (invariant + drift checks) that now runs periodically

Projects

Vision-based Autonomous Driving System | *Python, OpenCV, TF Lite, Raspberry Pi* — June 2025 – Present

- Built a Raspberry Pi RC car that follows lane markings in real time with an OpenCV lane-detection pipeline and a PID-style steering controller
- Trained a lightweight TF Lite (Edge Impulse FOMO) detector for stop signs and traffic lights that overrides steering to slow down or stop

Multiplayer 3D Chess with Hand Tracking | *Python, OpenCV, MediaPipe, Unity, C#* — Apr. 2026 – May 2026

- Built an online chess game played with hand gestures: MediaPipe tracks the thumb as a cursor and a pinch moves pieces; Python streams tracking to Unity over TCP, with Mirror syncing two players

Nightly – On-device Sleep Analyzer | *Swift, SwiftUI, SwiftData, HealthKit* — July 2026 – Present

- Private iOS app that records sounds overnight, transcribes speech on-device, and syncs to Apple Health

Web Chatroom Application | *Python, MySQL, MongoDB* — June 2025

- Real-time web chatroom for a Database Systems course; designed the SQL schema and compared MySQL vs. MongoDB storage

Courses & Certifications

NVIDIA DLI Certificates | *Fundamentals of Deep Learning, LLM Prompt Engineering, RAG Agents* — Nov. 2025
Introduction to Data Science, Penguin Academy | *Pandas, Matplotlib, TensorFlow, Keras, PyTorch* — Aug. 2024